

Poros: Perception-Enabled Interactive Assistance for Appliance Use by Blind and Low Vision Users

Yuxin Shen^{*} , Anqi Li^{*} , Wenxuan Zhao , Wei Liang[†] 

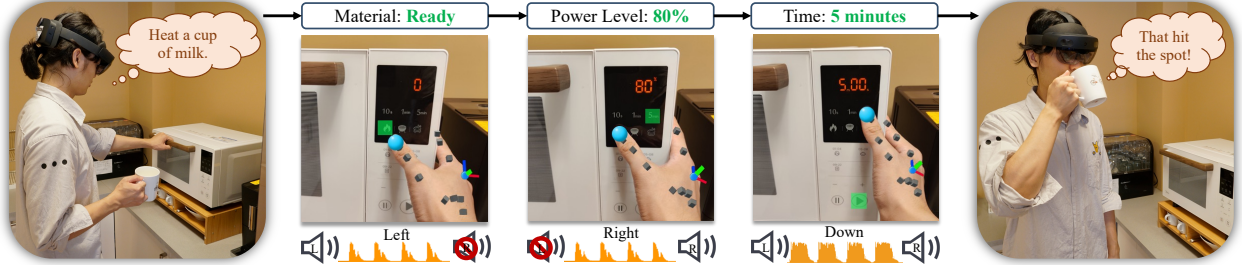


Figure 1: **Poros, a perception-enabled interactive assistance system for blind and low vision (BLV) users.** Poros helps BLV users operate appliances independently through real-time perception and interactive guidance. In this example, after receiving a voice command “heat a cup of milk”, Poros continuously tracks the user’s hand movements and infers the microwave state, such as material selection, power level, and time setting. It then provides step-by-step voice-based assistance and error correction until task completion.

Abstract—Modern home appliances pose significant accessibility barriers for individuals with visual impairments, driven by increasing functional complexity and minimalist design trends. Drawing on formative interviews with blind and low vision (BLV) participants, we propose Poros, a novel system providing perception-enabled interactive assistance to facilitate seamless appliance interaction. Our approach features an automated instruction generator that models appliance operation logic directly from user manuals to synthesize executable guidance for arbitrary tasks. Integrated into an Augmented Reality (AR) wearable, Poros provides real-time auditory feedback synchronized with the user’s hand gestures and the appliance’s display state. A user study with quantitative analysis and qualitative insights shows that our approach enhances user independence, enabling the successful execution of complex, multi-step operations that were previously considered inaccessible.

Index Terms—Accessibility, task guidance, blind and low vision, large language model

1 INTRODUCTION

Home appliances have transformed modern living by automating domestic labor and improving hygiene standards. However, while sighted individuals can easily navigate these systems, blind and low vision (BLV) users often encounter accessibility barriers [1, 25]. Many appliances rely heavily on visual indicators, such as LED displays and status lights, to convey system states and operational feedback, which are inaccessible to BLV users [17]. In pursuit of minimalist aesthetics, contemporary appliances increasingly replace physical buttons with flat touchscreens [18], eliminating the tactile cues that BLV users rely on to locate and operate controls. Furthermore, the rapid expansion of appliance functionalities has increased operational complexity, making interactions harder to understand and navigate for BLV users.

Prior work has explored multiple approaches to improving appliance accessibility for BLV users. Early approaches augment touchscreens with tactile overlays [18, 20, 23], or enable users to control appliances through smartphones or voice assistants [2, 32, 44, 50]. However, these solutions often require additional hardware or installation costs and are

difficult to generalize to complex appliances. Another line of research focuses on helping BLV users perceive appliance interfaces. Prior work has applied computer vision techniques to identify appliance categories [43], read display labels [14, 16], or localize control components [15, 28]. More recent work further integrates Large Language Model (LLM) to generate contextual visual descriptions and support real-time queries [8, 34]. To address the lack of tactile cues, some systems use hardware proxies that physically interact with appliance interfaces on users’ behalf [26, 27]. Although these approaches enhance BLV users’ comprehension of interface layouts and device statuses, they provide limited guidance for completing operational procedures.

To facilitate understanding of operational procedures, recent work has also explored systems that support users in following multi-step tasks [22, 31, 51]. For example, InstruMentAR automatically generates AR tutorials from recorded user demonstrations [31], while Guided Reality produces embedded and dynamic AR guidance based on structured step-by-step instructions [51]. Specifically for the BLV community, Vid2Coach transforms how-to videos into wearable camera-based assistance that provides mixed-initiative feedback during task execution [22]. Despite these advancements, a critical gap remains: existing systems do not fully address the nuanced challenges of interacting with physical appliance interfaces. Operating household appliances requires users to locate specific components, interpret dynamic device states, and translate abstract procedural instructions into concrete physical operations, which remains profoundly challenging for BLV users [19].

Drawing on insights from prior work on assistive perception and procedural guidance systems, as well as formative interviews with nine BLV participants, we propose Poros, a perception-enabled assistance system that supports BLV users in operating a wide range of household appliances throughout the entire task process in real time. Perception-enabled refers to the use of visual perception from AR headset cameras

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to continuously sense and interpret user hand movements and appliance display states for context-aware task assistance. Poros includes an instruction generator that models operational procedures from user manuals and produces task-specific action sequences. The system is deployed on a wearable HoloLens AR headset. Using the headset's built-in camera, Poros applies computer vision techniques to track the user's hand position and recognize appliance states by interpreting display information on the appliance interface. Based on the perceived user actions and appliance states, the generated operation sequence is dynamically translated into executable speech instructions that guide the user step by step. Figure 1 illustrates an example scenario in which a BLV user heats milk in a microwave under the guidance of Poros.

Unlike traditional instruction manuals that provide abstract textual descriptions (e.g., setting the time to five minutes), Poros converts operational rules into fine-grained and executable action guidance based on the user's ongoing actions, appliance states, and task models. This enables BLV users to operate appliances without relying on instructions or memorizing complex procedures, allowing them to independently explore advanced appliance functions. We evaluate Poros through a comparative user study with quantitative and qualitative analyses across tasks of varying complexity. The results demonstrate that our approach significantly enhances user independence and enables BLV users to successfully perform complex multi-step appliance operations that were previously difficult or impossible to accomplish unaided. Our findings also reveal design implications and future directions for appliance-assistive systems, including adaptable guidance granularity, customizable auditory feedback, and diverse interaction styles.

In summary, this paper makes the following contributions:

- We design Poros, a perception-enabled assistance system that provides real-time procedural guidance to support BLV users in operating household appliances independently.
- We introduce an instruction generation framework that models operational logic from household appliance manuals, generates executable step-by-step instructions, and iteratively refines guidance through interactive exploration.
- We evaluate the proposed system through a comparative user study that incorporates both quantitative analysis and qualitative feedback, demonstrating its effectiveness in enhancing BLV users' independence and improving their ability to operate appliances, particularly for complex tasks.

2 RELATED WORKS

2.1 Assistive Technologies for BLV Appliance Use

Prior work has explored appliance accessibility for BLV users through interface augmentation, smart home control, and vision-based interface perception. Interface augmentation adds tactile overlays or 3D-printed templates to touchscreen appliances, helping blind users locate and distinguish interface elements through touch [18, 20, 23]. However, these methods require appliance-specific customization and scale poorly to complex or dynamic interfaces. Smart home approaches allow BLV users to operate appliances via smartphones or voice assistants [2, 13, 32, 47, 50]. While they reduce the need to locate physical buttons, they provide limited feedback and depend on smart devices or infrastructure.

Recent accessibility research has focused on vision-based interface perception. Prior work has used computer vision to recognize appliance categories [14, 43], access display content [16, 17, 19], and localize controls on appliance interfaces [15, 28]. For example, StateLens provides interactive guidance to help blind users access dynamic touchscreen interfaces [19]. To facilitate interface understanding, recent work combines computer vision with LLMs to generate contextual visual descriptions and support real-time environmental queries, such as WorldScribe and CustomSight [8, 34]. Commercial systems such as Be My Eyes/AI¹ also provide visual assistance through AI-generated scene descriptions. While these vision-based and scene-description systems improve access to visual information through interface perception, image analysis, or scene descriptions, they offer limited support for

modeling appliance operation workflows. In contrast, our work adopts a top-down strategy that constructs the operation space from instruction manuals and models appliance operation logic as multi-step procedures with fine-grained operations grounded to concrete buttons at each step.

2.2 Instruction Understanding and Task Guidance

Instructions help users understand task procedures across domains such as appliance operation, software usage, and game interaction. Prior work has explored extracting procedural knowledge from structured visual manuals [38, 45, 46] and natural language instructions [30, 48]. For example, Shao et al. [38] reconstruct furniture components and assembly processes from assembly manuals, while Wang et al. [45] parse visual manuals into machine-interpretable assembly instructions. Other work focuses on procedural logic in natural language instructions. Liu et al. [30] extract task-relevant information from ambiguous instructions through task-centered semantic analysis, and Wu et al. [48] enable agents to learn Atari games from textual manuals.

Recent research has explored systems for multi-step task assistance during real-world execution. Several works use AR or multimodal sensing to provide situated guidance [9, 22, 31, 51]. For example, Liu et al. [31] generate AR tutorials from recorded demonstrations, Chen et al. [9] transform paper instructions into spatialized AR experiences, and Zhao et al. [51] generate spatially embedded AR guidance from structured task descriptions. However, these systems mainly support sighted users. Research for BLV users remains limited and often targets simple activities; for instance, Huh et al. [22] transform how-to videos into wearable camera-based assistants that provide mixed-initiative feedback and guidance. Although such systems improve accessibility for specific activities, they do not fully address complex physical interfaces. Operating household appliances requires users to locate controls, interpret dynamic states, and translate instructions into physical actions, which remains challenging for BLV users [19]. Our work addresses this gap by modeling appliance logic from manuals and generating interaction-aware instructions aligned with users' real-time actions.

2.3 Interactive Assistance Systems

Lightweight consumer-grade smart devices have enabled new forms of interactive guidance. AR is a promising platform for situated assistance and has been widely applied in domains such as assembly guidance [37], system configuration [33], and equipment maintenance [5]. AR has also been used in assistive technologies for BLV users. Early work introduced auditory AR environments that integrate acoustic virtual objects into real-world scenes to convey spatial information [36]. Later studies applied AR headsets to support obstacle avoidance and navigation, typically using audio or haptic feedback to enhance spatial awareness [11]. For example, Hu et al. [21] investigated scene sonification strategies that convert environmental information into auditory cues for blind users. Moreover, SeeingVR [52] and EnVisionVR [10] improve visual accessibility in Virtual Reality (VR) by interpreting virtual scenes and delivering feedback through auditory, haptic, or multimodal cues.

Wearable devices have also been explored as platforms for interactive assistance. Wearable interfaces enhance access to visual information through interaction techniques, such as ring-based input for smartphone operation [29] or on-body interaction [12, 35, 42]. Systems such as FingerReader [3] and AiSee [4] integrate optical character recognition with wearable interfaces to support activities such as grocery shopping. Similarly, navigation-oriented systems explore multimodal feedback for independent navigation [49]. Building on these sensing and interaction capabilities, we use commercial AR headsets to build an interactive assistance system for appliance operation. Compared with systems focused on navigation, object recognition, or general task guidance, our system supports BLV users in interacting with complex appliance interfaces and completing multi-step operational procedures.

3 FORMATIVE STUDY

Before developing our system, we conducted a formative study using semi-structured interviews to understand how BLV users interact with household appliances and the challenges they encounter. The insights from this study informed the design of our system.

¹<https://www.bemyeyes.com/>

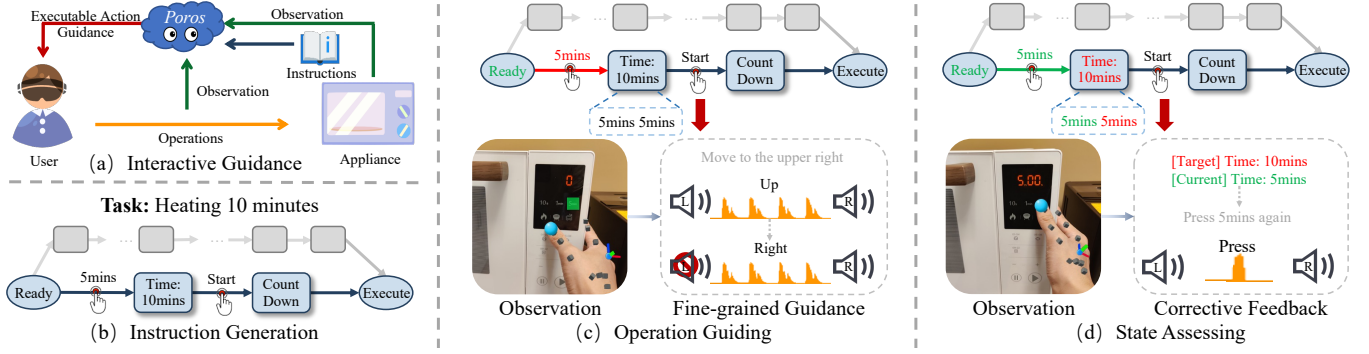


Figure 2: **Overview of the Poros system.** (a) Given an instruction manual, Poros provides interactive guidance. (b) For a given task, it generates an instruction sequence. (c) It then provides audio guidance while monitoring hand movements and the appliance display to support user operations. (d) When a step is completed (green), Poros assesses the current state against the target state to determine the next action (red).

Participants We recruited 9 BLV participants through local accessibility communities and online groups (5 females, 4 males), aged 21–36 years ($M = 27.2$, $SD = 4.82$). The participants were recruited from different regions in Asia and represented diverse occupational backgrounds, including students, musicians, actors, corporate employees, and freelancers. In terms of visual conditions, 3 participants had Congenital Total Blindness (CTB), 5 had Acquired Total Blindness (ATB), and 1 had Congenital Blindness with Light or Color Perception (CB-LC), while none had Acquired Blindness with Light or Color Perception (AB-LC). These categories characterize participants’ diverse visual experiences and align with prior accessibility research on BLV users [41]. They situate our study within the broader BLV population, including tens of millions of people with blindness and hundreds of millions with moderate or severe vision impairment worldwide [6].

Procedures We asked participants about their experiences with household appliances, including the types and frequency of appliances they commonly use in daily life, how they learn to operate new appliances, and the difficulties they encounter when using them. To further probe these challenges, we presented several common appliances (e.g., a refrigerator and a microwave oven) and asked participants to describe the typical steps they would take to operate them. Each interview lasted about 30 minutes, and participants were compensated with \$20. Detailed interview questions are provided in the supplementary material.

Data Analysis We analyzed the interview transcripts and open-ended responses using inductive thematic analysis [7]. The first two authors independently reviewed the data and conducted open coding to identify recurring statements related to appliance-use contexts, assistive strategies, feedback mechanisms, interface difficulties, and expectations for appliance functions and controls. The initial codes were compared and refined through iterative discussion, with disagreements resolved by revisiting the transcripts and reaching consensus. The other two authors reviewed the resulting codes and themes, and all authors finalized the thematic framework used to organize the formative-study findings.

Findings The findings are organized into three themes: overall usage, auxiliary feedback, and design implications.

Overall Usage. Participants identified two main categories of household appliances: chore-related and entertainment appliances. Microwave ovens were the most commonly used appliances (7 participants, 77.8%), followed by refrigerators and washing machines (5 participants). Six participants (66.7%) considered air conditioners essential despite not using them daily, whereas entertainment appliances such as televisions were used less frequently, often with family or friends. Only one participant reported regularly using small appliances such as electric kettles or air fryers. When using appliances, seven participants (77.8%) reported feeling uncertain when first encountering a new appliance. A major challenge was completing tasks that required multiple sequential operations or repeatedly pressing multi-mode buttons on the interface. Four participants specifically mentioned frustration when operating multifunctional digital appliances. As a result, participants

adopted coping strategies such as asking others for help (4 participants), placing markers on operation panels (5 participants), or using smartphone applications for smart appliances (4 participants). Many participants, therefore, relied only on simple functions and avoided complex features due to the higher risk of errors. We further observed descriptive patterns across participants’ vision-impairment types. Participants with CTB more often emphasized tactile exploration and physical markers, while participants with ATB more often described relying on prior visual memories or help from sighted people when learning unfamiliar appliances. Five participants expressed a need for assistive tools to guide or supervise appliance operation. For instance, P06 commented on repeated operations: “*Tasks tend to succeed only when the correct buttons are pressed in order, but I would never know whether I pressed the right button unless I tasted the food.*” Three participants also reported that function prompts could prevent them from abandoning certain features due to forgotten buttons or steps.

Auxiliary Feedback. Participants highlighted limitations in existing auxiliary feedback mechanisms, including tactile and auditory cues. For example, P01 described difficulties with microwave interfaces: “*The microwave touchpad is sometimes not touch-sensitive. It is hard for me to locate the specific buttons.*” Participants who placed markers on appliance panels mainly used simple stickers to locate buttons rather than to indicate specific functions. Notably, none of the participants labeled appliance buttons with Braille. Five participants could not read Braille, while the others considered simple stickers more practical due to the limited space on appliance panels. As P09 explained, “*There is not enough space to paste button functions written in Braille.*” Auditory feedback, typically in the form of beeping sounds, was also insufficient. Participants found these sounds too abstract to convey meaningful information, making it difficult to determine which function had been triggered or whether the correct button had been pressed.

Study Outcomes. These findings indicate that interacting with household appliances can be challenging, particularly when operations involve multi-step procedures and multi-mode button interactions, with no clear prompts for appliance states and functions. This highlights the need for explicit, step-by-step guidance that helps users understand appliance interfaces and operation procedures. Based on this insight, we use kitchen appliances as a representative scenario and design a system that provides real-time guidance grounded in users’ interactions with appliance interfaces to support BLV users. Furthermore, participants’ existing appliance-use strategies informed the baseline conditions in our comparative experiments to evaluate the proposed system.

4 POROS SYSTEM

We design Poros, an interactive assistance system that enables BLV users to operate appliances independently. Poros models appliance usage from instruction manuals and generates step-by-step operation instructions for user-specified tasks (Fig. 2(a)). Users activate the system via voice commands (e.g., “heat with 80% power”). During task execution, Poros guides users through the required operations

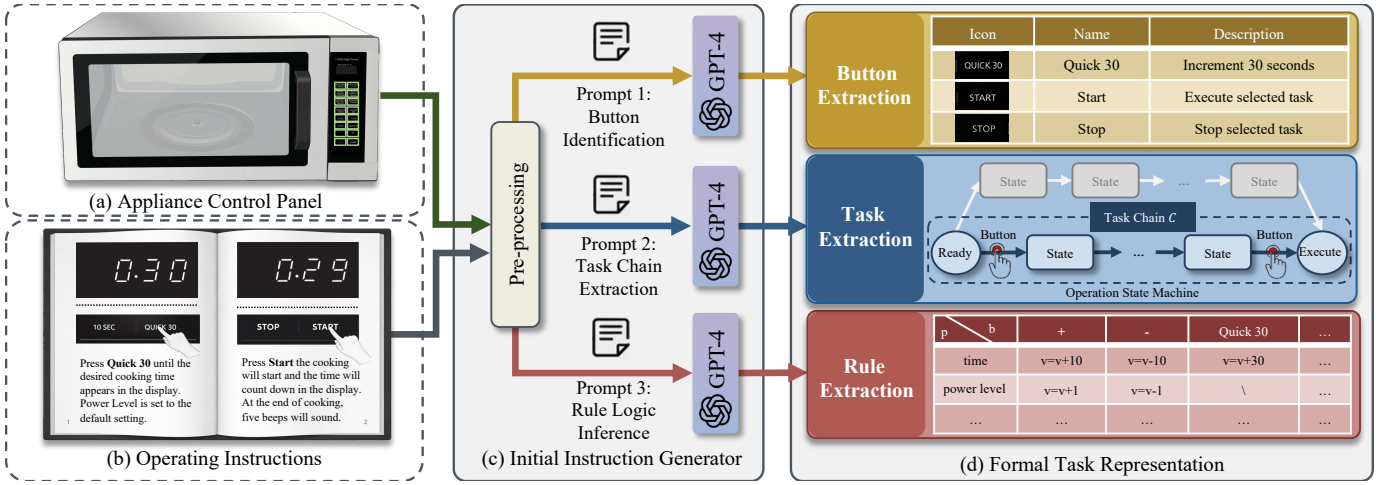


Figure 3: **Overview of the Initial Instruction Generator.** Given the appliance control panel layout (a) and the textual operating instructions (b), the system applies an LLM-based extraction pipeline (c) to construct an operation state machine representation of the appliance’s behavior. The pipeline consists of three stages: identifying interactive buttons from the control panel, extracting task sequences from the instructions, and inferring rule-based logic for parameter updates and state transitions. The resulting Formal Task Representation (d) encodes the appliance’s operational logic and serves as the basis for downstream instruction generation and execution guidance.

while monitoring both the appliance interface and the user’s hand movements. By observing the appliance state, the system verifies whether each operation has been completed and determines the next instruction accordingly. This closed-loop interaction enables Poros to continuously provide step-by-step guidance until the task is completed.

Poros consists of two main components: an **instruction generator** and an **interactive guidance module**. The instruction generator parses appliance manuals to learn the operational rules and feasible state transitions of the appliance, allowing the system to generate an instruction sequence for a specific task (Fig. 2(b)). This sequence represents the ordered set of button operations required to reach the target appliance state. For example, when heating food using a microwave oven, the generated instructions may include steps such as “press the 5-minute setting button twice” followed by “press the start button once.” Based on the generated instruction sequence, the interactive guidance module assists users in executing the operations. The system continuously monitors the user’s hand movements and the appliance interface via the headset camera (Fig. 2(c)). According to the user’s current hand position, Poros provides auditory guidance to direct the next action, such as moving the hand toward a specific button or pressing it. Meanwhile, Poros observes the appliance interface to assess whether the intended operation has been successfully executed (Fig. 2(d)). For example, the system checks whether the display shows the expected time setting (e.g., 10 minutes). Based on this feedback, Poros decides whether to proceed to the next instruction or provide corrective guidance.

In this work, we use a microwave oven as a representative example to demonstrate the proposed pipeline. The framework can be extended to other appliances with structured multi-step operations.

4.1 Instruction Generation

When sighted users operate a new appliance, they typically consult the instruction manual to understand the operation process before interacting with the device. For example, when heating food using a microwave oven, users form a rough workflow (e.g., setting the heating power and then the heating time) and then translate this workflow into a sequence of concrete actions, such as pressing the “power level” button twice, pressing the “1 min” button three times, and finally pressing “start”. These actions are executed sequentially to achieve the desired appliance state. However, according to our formative study, BLV users often struggle to use new appliances or access advanced functions because they lack knowledge of the underlying operation process. To address this challenge, our goal is to automatically generate executable operation instructions that guide BLV users through appliance interactions.

A key challenge is that appliance manuals combine diagrams, textual descriptions, abstract rules (e.g., “add one minute with each press”), and example procedures. While users can easily generalize from these rules and examples, converting diverse descriptions into machine-readable instructions can be challenging. To address this issue, we model appliance operation as an **operation state machine**. In this representation, each state corresponds to a parameter-setting mode of the appliance (e.g., time or power), and button presses correspond to transitions that modify the appliance’s parameters. This formulation allows complex appliance operations to be represented as sequences of state transitions, enabling systematic generation of operation instructions for arbitrary tasks. To automatically construct this model, we leverage the reasoning capabilities of LLMs to interpret appliance manuals and extract the underlying operational logic, including buttons, parameters, and their transition rules. Based on these extracted elements, we build an initial instruction generator that produces executable instruction sequences for user-specified tasks, as shown in Fig. 3. Inspired by how humans refine their understanding of a device through trial and exploration, we introduce an interactive validation process to verify frequently used instructions and refine the generator based on appliance feedback.

4.1.1 Instruction Representation

To formalize appliance operations, we represent the appliance control logic as an operation state machine. In this model, appliance usage is expressed as transitions between parameter-setting states triggered by button presses. An appliance typically contains several configurable parameters p , corresponding to different operation modes (e.g., time, power level, weight, or temperature). We denote the set of appliance parameters as P , where each parameter represents a controllable attribute of the appliance. The button set B includes all buttons on the appliance control panel. A state represents the process of configuring a specific parameter and is defined as

$$S = (p, v, A),$$

where $p \in P$ denotes the parameter being configured, v represents its current value, and $A = \{a_i \mid a_i \in B\}$ denotes the set of action buttons that can modify the parameter value in this state.

Each action button a modifies the parameter value according to a rule function f . These rule functions describe how button operations update appliance parameters and collectively form the rule function set F . Formally, each rule is represented as a function describing how a button updates a parameter value:

$$f_{p,b}(v) = v + r_{p,b},$$

where p denotes the parameter, b denotes the action button, and $r_{p,b}$ represents the value change of parameter p triggered by button b .

In addition to parameter-setting actions, specific buttons function as transition buttons. A transition button $t \in B$ triggers a shift between appliance states. Building on these elements, an operation process from the manual can be defined as a **task chain** C , which represents a path in the operation state machine: $C = (S_1, t_1, S_2, t_2, \dots)$. This representation models appliance operation as a sequence of state transitions, providing a structured basis for generating executable instruction sequences.

4.1.2 Initial Instruction Generator

In this phase, we construct an initial instruction generator from appliance instruction manuals by extracting the structural elements required for the operation state machine. The instruction manuals were prepared as digital text and used as input to the generator.

Button Extraction We first identify the set of buttons on the appliance control panel. The textual content from the control panel interface (as shown in Fig. 3(a)) is obtained using OCR-based recognition and provided to GPT-4, an LLM. Given a task-specific prompt, the LLM performs semantic grounding over these contents to interpret their functional meanings in the context of the appliance interface and identify interactive buttons. Non-interactive elements, such as display regions or decorative icons, are filtered out. The remaining labels are mapped to a set of button identifiers, forming the button set B , which serves as the atomic interface representation for subsequent task modeling.

Task Extraction We extract the procedural operations outlined in the instruction manual. Each procedure is transformed into a task chain that includes parameter-setting states and transition buttons, capturing both the sequential operations and state transitions of appliance usage. To achieve this, we prompt LLMs to analyze the textual procedure and perform two tasks: (1) identify the sequence of buttons mentioned in the procedure, and (2) determine which parameter each button modifies. The extracted elements are organized using a predefined format. Buttons that modify the same parameter are represented as `{button_set}: parameter`, while standalone buttons are represented as transition buttons. For instance, the operation process shown in Fig. 3(b) can be represented as: (i) `{Quick 30}: Time`; (ii) `Start`. In this representation, buttons that are grouped with a parameter create a state S . The action button set A contains the grouped buttons and the parameter value v is initialized to 0. Buttons that are not associated with a parameter correspond to transition buttons T . States and transitions are then organized sequentially to form a task chain representing the operation procedure. Because LLMs output may contain inconsistencies due to ambiguity in procedural language and variability in appliance manuals, we apply a lightweight rule-based post-processing step to refine the generated task chains and improve structural consistency. If a transition button also appears in the action button set of the adjacent state, the transition is absorbed into that state. Additionally, if two adjacent states correspond to the same parameter and no transition button exists between them, the states are merged, and their action button sets are combined. Using this approach, we generated task chains for 59 operation procedures from nine appliance manuals, achieving an extraction accuracy of 89.8%. The nine appliance manuals include commercial microwave ovens and micro-baking integrated machines from multiple manufacturers (e.g., Midea and Panasonic), covering a range of real-world appliance interfaces. Detailed results are provided in the supplementary material.

Rule Extraction Finally, we extract the parameter update rules associated with each action button. To accomplish this, we provide the LLMs with the task description, the parameter name p , the button label a_i , and its description from the instruction manual. Based on this information, the LLMs infer how pressing the button modifies the parameter value. Each rule is represented in the form $p * a_i$, indicating that pressing button a_i transforms the current value of parameter p into a new value. This representation can capture different types of parameter updates, such as increasing or decreasing a value, assigning a predefined value, or cycling through a set of options, depending on the appliance manual. For example, the rule for the button *Quick 30*

can be represented as: $f_{\text{time,Quick } 30}(v) = v + 30$. All extracted rules form the rule set of the instruction generator.

4.1.3 Refined Instruction Generator

Since the instruction generator is constructed automatically by prompting LLMs with the appliance manual, the generated rules may not always match the behavior of the appliance. Inspired by how humans refine their understanding of a device through exploration, we introduce a validation strategy to refine the initial generator. This validation process checks whether the generated operation state machine is consistent with the actual appliance behavior. Specifically, we first prompt LLMs to generate 500 candidate tasks. The initial instruction generator produces button sequences for these tasks. The generated button sequences are executed on the appliance, and the observed appliance feedback is compared with the predicted state to verify whether the corresponding rules and transitions are correct. To improve efficiency, we prioritize validating the most frequently used rules. We assess the frequency of each rule and select those with a cumulative frequency exceeding a predefined threshold. In our microwave oven example, the threshold is set to 80%. We further analyze the relationship between the threshold value and the number of required validation operations to understand the validation cost. Detailed results are provided in the supplementary material. For each selected rule, we identify the shortest task chain containing the rule and construct a task that triggers the rule exactly once. The system then generates the corresponding button sequence and asks the user to execute it. If the appliance feedback matches the predicted outcome, the rule is considered validated; otherwise, it is corrected based on the observed feedback. After validation, the refined rule set is used by the instruction generator to generate operation instructions, ensuring they are consistent with the appliance's behavior.

4.1.4 Instruction Sequence Generation

Given a task specifying the target values of appliance parameters, Poros generates the corresponding sequence of button presses. The system first selects a task chain whose states correspond to the parameters specified in the task. In each state, the parameter value is initialized to 0, while the task determines the target value. Each button updates the parameter value according to the rule. We then employ dynamic programming to compute the sequence of button presses that reaches the target value with the minimum number of operations. The final instruction sequence is obtained by combining the transition buttons and the action sequences across states.

4.2 Interactive Guidance

Although the generated instructions describe the desired operations, executing them remains challenging for BLV users because appliance interactions rely heavily on visual information, such as locating buttons and observing device feedback. Therefore, Poros provides interactive guidance throughout the entire task execution process through two components: **operation guiding**, which helps users locate and press the correct buttons, and **state assessing**, which enables users to perceive the appliance's current state and determine the next action. We also measured the end-to-end processing latency from receiving a camera frame to triggering audio feedback, which averaged approximately 75 ms in our prototype, supporting responsive real-time interaction.

4.2.1 Operation Guiding

Operating household appliances typically involves physical interactions such as opening doors or pressing control buttons. To help users follow the generated instructions accurately, Poros continuously guides the user's hand toward the target button by minimizing the spatial distance between the user's hand and the target button until the hand reaches the corresponding operation area. We obtain the 3D position of the user's hand using the HoloLens's built-in hand-tracking functionality. Specifically, we track the top joints of the index finger and the thumb, as these are the most commonly used fingers for pressing buttons. Users can choose either finger based on preference during operation. During appliance-specific setup, we manually annotate the control panel by marking each physical button as a 2D region and assigning its label

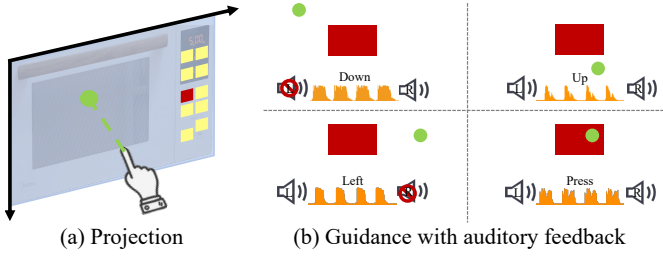


Figure 4: **Operation guiding.** (a) The hand position (green circle) is projected onto the microwave panel surface. A virtual trigger region is defined for each physical button (red and yellow rectangles). (b) The relative position between the green circle and the target button (red rectangle) is converted into auditory guidance.

and function. Crowdsourcing-based GUI labeling approaches could potentially automate this manual annotation step in future systems. The center of each region is used as the button location in the panel coordinate system. At runtime, Poros registers the appliance panel using an AR marker technique (e.g., the Vuforia AR toolkit). When the HoloLens camera captures the predefined marker, the system obtains the panel’s spatial pose and maps the annotated button locations from the panel coordinate system to the HoloLens coordinate system. In our prototype, the microwave oven’s front surface serves as the AR marker.

During operation, Poros computes the spatial difference between the user’s hand and the target button and converts it into auditory feedback to guide the user toward the correct location. As shown in Fig. 4(a), the hand position (e.g., the top joint of the index finger) is vertically projected onto the microwave panel. The relative position between the hand and button is then computed along four directions: Left, Right, Up, and Down, and directional audio feedback guides the user’s movement. For example, as shown in the top-left of Fig. 4(b), if the user’s finger is located at the upper-left of the target button, the system provides the audio instruction “Down” to guide the user toward the correct direction. To determine when the user reaches the target button, we define a virtual trigger region for each physical button, as shown in Fig. 4(a). When the user’s finger enters this region, the trigger is activated and Poros prompts the user via speech to press the button (bottom-right of Fig. 4(b)). To ensure robustness, the trigger region is implemented as a square with a width of 0.5 cm centered on the button. The success of the button press is verified by the state assessment described in Sec. 4.2.2.

4.2.2 State Assessing

BLV users also need to understand the appliance’s current state to determine the next operation. Because appliance states are typically conveyed through visual displays, such as numerical indicators, users may find it difficult to verify whether an operation has been successfully executed. To address this challenge, Poros uses a vision-based display-state recognition pipeline to convert camera observations of the appliance display into symbolic appliance states and provide feedback to users. Using the microwave oven as an example, each button press updates the appliance state displayed on the screen, such as the heating duration or power level. Since many appliances use LED displays for numeric feedback, we design a lightweight pipeline to recognize displayed numbers and infer the current state. The pipeline consists of three stages: segmentation, recognition, and assessment.

Segmentation. The process begins by dividing the captured image into sub-images, each containing a single displayed number. The HoloLens camera captures images at a fixed 200-millisecond interval. The captured image is preprocessed using image processing techniques, including edge detection to extract foreground regions and morphological closing to merge nearby connected regions. A bounding rectangle is computed for each connected region to obtain the segmented digits.

Recognition. Next, each segmented region is represented using Histogram of Oriented Gradients (HoG) features and passed to a pre-trained classifier to predict the corresponding digit label. The classifier is implemented using an off-the-shelf Support Vector Machine (SVM).

Since recognition is performed in a relatively controlled indoor environment with limited variation in orientation, scale, and illumination, the method provides reliable digit recognition for our prototype system.

Assessment. Finally, the recognized appliance state is compared with the desired state in the generated instructions. If the recognized state matches the expected state, the system proceeds to guide the user to the next operation step. Otherwise, corrective feedback is provided, such as prompting the user to repeat the previous action. The corrective instruction is determined by computing the shortest transition path from the current state to the target state based on the instruction generator.

5 USER STUDY

To evaluate the effectiveness of our approach from the perspective of BLV users, we conducted a comparative user study to investigate whether Poros can assist BLV users in operating household appliances. We compared Poros with two baseline approaches:

1) Conventional: Participants operated the microwave using its original interface, relying on the tactile feedback of the touchpad and the beep produced when pressing a button.

2) Marker: Raised tactile markers were attached to each button surface. In this condition, uniform circular markers were used instead of Braille labels, following prior studies [24].

5.1 Experiment Settings

Participants We recruited 14 BLV participants through local accessibility communities and online groups (5 females and 9 males), whose ages ranged from 21 to 36 years ($M = 26.8$, $SD = 4.37$). Their visual conditions comprised 5 CTB, 5 ATB, 3 CB-LC, and 1 AB-LC. Each participant received \$40 as compensation.

Experimental Setup We used a common microwave oven, Midea PC2322W, with 14 touch-sensitive buttons for time adjustment, power level control, predefined menus, cancellation, and confirmation. The buttons are nearly flat and provide minimal tactile feedback. The study appliance was one of the nine appliances included in the instruction-generation evaluation, and its operation state machine was generated by the instruction-generation pipeline and validated on the physical appliance before the participant sessions. Participants interacted with the microwave under three conditions: Poros, Marker, and Conventional. Based on formative study insights, we designed two task difficulty levels. Simple-Level tasks involved heating food with specified time and power settings, while Hard-Level tasks required users to prepare dishes using predefined menu functions with customized parameter settings. We created four tasks for each level, with tasks at the same level using similar operations, button types, and numbers of button presses. To minimize learning effects, we randomized the order of the three assistance conditions and distributed the six possible orders as evenly as possible across participants. We randomly assigned tasks to approaches within each difficulty level to reduce task-specific learning effects. Task descriptions are provided in the supplementary materials.

Procedure Before the study began, participants were informed about the procedures and provided informed consent. The experimenter familiarized participants with the microwave by verbally introducing the control panel and button functions without exposing them to experimental tasks. Participants could ask questions, but they did not touch the microwave, operate buttons, or practice task sequences. Task-specific voice commands were provided only at the start of each Poros trial. For each difficulty level, tasks were randomly selected from the predefined task set and assigned to one of the three approaches (Poros, Marker, and Conventional), ensuring that each participant experienced all approaches. Before the participant sessions, the experimenter prepared Poros for the study appliance by obtaining the digital manual, constructing the manual-based model, annotating button regions and functions, registering the appliance panel to the HoloLens coordinate system, and validating selected operation rules. This setup was reused across participants and tasks, so participants did not perform appliance-specific setup or manual uploading. When using Poros, the experimenter initialized the system’s listening mode before each trial, and participants then started the assigned task by speaking the task command. The system

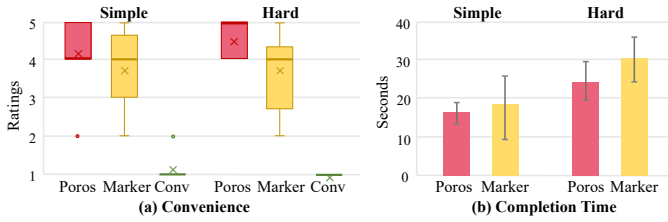


Figure 5: **Results on subjective convenience ratings and completion time.** (a) Convenience ratings for Poros, Marker, and Conventional (Conv) methods under Simple and Hard tasks. (b) Completion time for Poros and Marker. Error bars indicate standard deviation.

used the built-in HoloLens speech recognition API to interpret speech commands and convert them into task representations consisting of state-parameter pairs. Based on these commands, Poros generated step-by-step operation instructions for participants to follow. Participants were allowed to ask for help with task procedures, but we did not assist with button locations or the microwave’s current state.

After completing each task, participants rated convenience and confidence. Convenience measures how easy it was to operate the microwave, while confidence reflects how certain participants felt about the correctness of their operations. Ratings were provided on a 5-point Likert scale, with 1 indicating low convenience or confidence and 5 indicating high convenience or confidence. We also recorded help requests, task completion time, and task success rate. A help request was counted when a participant asked for task-procedure assistance, such as what operation should be performed next or how many times an operation should be repeated. Requests about button locations, button identities, or the microwave’s current state were excluded, and the experimenter did not provide such assistance. Task completion time was measured from the task command to task completion. A task was successful when the participant completed the assigned task and reached the target appliance state, regardless of intermediate errors. Finally, we conducted post-study interviews to assess satisfaction with the three approaches and collect open-ended comments for contextualizing the results (see supplementary materials for detailed questions).

5.2 Results and Analysis

We analyzed the results of our user study guided by the following research questions: Did participants find our appliance guidance system helpful? If so, in what ways does the system support BLV people? In the following, we summarize the key observations from the study. Four participants were unable to complete the full study due to technical issues (e.g., HoloLens overheating, sudden power outages, and microwave malfunction). Data from these incomplete sessions were excluded from the statistical analysis and hypothesis testing, although their interview responses were retained for qualitative analysis. As a result, the analysis and reported metrics were computed based on the valid trials, including data from 12 participants for the simple-level tasks and 10 participants for the hard-level tasks. For each condition, participants completed three tasks, providing three data points per participant.

Poros provides convenient support for microwave operation. One key design goal of Poros is to provide convenient support for BLV users during appliance operation. We collected participants’ ratings of perceived convenience after each task, which reflected participants’ perceived ease of operating the microwave under each approach. We also measured task completion time to provide an objective comparison.

Convenience Ratings. Poros received the highest convenience ratings among the three approaches for simple-level and hard-level tasks (Figure 5(a)). For Simple tasks, Poros received a median convenience rating of 4 (Md = 4), similar to Marker (Md = 4) but higher than Conventional (Md = 1). For hard tasks, Poros achieved the highest median rating (Md = 5), followed by Marker (Md = 4) and Conventional (Md = 1). We conducted Friedman tests on convenience ratings across the three approaches for Simple and Hard tasks. The results showed significant differences among the approaches in both Simple

($\chi^2(2) = 20.36, p < .05$) and Hard ($\chi^2(2) = 17.68, p < .05$) conditions. We performed pairwise Wilcoxon signed-rank tests with Bonferroni correction ($\alpha = .017$) as post-hoc analysis. For Simple tasks, Poros achieved significantly higher ratings than Conventional ($Z = -3.10, p < .017$). Similarly, for Hard tasks, Poros significantly outperformed Conventional ($Z = -2.89, p < .017$). However, no significant difference was found between Poros and Marker in either task condition.

Qualitative feedback helped explain these quantitative results. In the Conventional condition, participants reported that the nearly flat touch buttons made target localization difficult. One participant noted, “For the conventional approach, the buttons are nearly the same height as the surrounding panel. I find it hard to locate the target button quickly, although I’m familiar with the button layout.” For the Marker condition, sensitivity to frequently used buttons, such as time and power controls, helped participants locate them easily with simple tactile markers. P13 stated, “Besides the buttons of enter and cancel, I only used markers on two function buttons: time and power level.”. However, hard-level tasks with nested menus and multi-step procedures caused cognitive strain because participants had to recall complex operation sequences. Poros was positively received in these scenarios because it reduced the need to learn and memorize operations. P06 noted, “I love it. I do not need to spend time learning and memorizing any operations. I just need to follow the speech guidance, and the dishes will be ready.”.

Task Completion Time. To evaluate task efficiency, we measured completion time as an objective metric, excluding failed trials to ensure a fair comparison. Due to the limited number of completed trials in the Conventional condition, this analysis exclusively compares Poros with Marker. As shown in Fig. 5(b), participants completed Simple tasks slightly faster with Poros ($M = 16.22, SD = 3.07$) than with Marker ($M = 18.00, SD = 8.60$). For Hard tasks, participants also completed tasks faster with Poros ($M = 24.86, SD = 5.05$) compared with Marker ($M = 30.29, SD = 5.79$). Overall, Hard tasks required more time than Simple tasks under both approaches. We conducted ANOVA tests on the completion time for Simple and Hard tasks separately. The results showed no significant difference between Poros and Marker in either Simple ($F_{1,8} = 0.34, p > .05$) or Hard ($F_{1,6} = 3.49, p > .05$) tasks.

Although the difference was not statistically significant, several patterns emerged. Participants located the “Enter” button the fastest (about 1.5 seconds) under both guidance approaches. This behavior may be explained by participants’ strategy of first identifying the location of the “Enter” button and then using it as a reference point to locate other buttons. Thus, the initial exploration positions were often close to this button. Although Poros occasionally required slightly more time than markers, participants generally considered the small time differences acceptable. As P12 commented, “It doesn’t bother me to take 1 to 2 more seconds to locate the button. Poros has already surprised me to help me operate the microwave in such an advanced way.”.

Poros helps users operate the microwave with confidence. Another key characteristic of Poros is that it provides real-time assistance throughout the operation process. We examined whether this assistance helped participants operate the appliance with confidence and complete tasks independently. After each task, participants rated their confidence in pressing the correct buttons and obtaining the correct results. We analyzed these ratings with help requests and task success.

Confidence Ratings. Confidence ratings from participants across the three approaches at both levels of the task are displayed in Fig. 6(a). For simple tasks, Poros received the highest median confidence rating (Md = 5), followed by Marker (Md = 4) and Conventional (Md = 1). For hard tasks, Poros again achieved the highest median rating (Md = 5), followed by Marker (Md = 3) and Conventional (Md = 1). To determine whether the differences were statistically significant, we performed Friedman tests on confidence ratings across the three approaches, evaluating Simple and Hard tasks separately. The results showed significant differences among the approaches in both Simple ($\chi^2(2) = 21.27, p < .05$) and Hard ($\chi^2(2) = 17.90, p < .05$) tasks. We further conducted Wilcoxon signed-rank tests with Bonferroni correction ($\alpha = .017$) as post-hoc analyses. For Simple tasks, Poros received significantly higher ratings than Conventional ($Z = -3.11, p < .017$), while the difference between Poros and Marker was not significant.

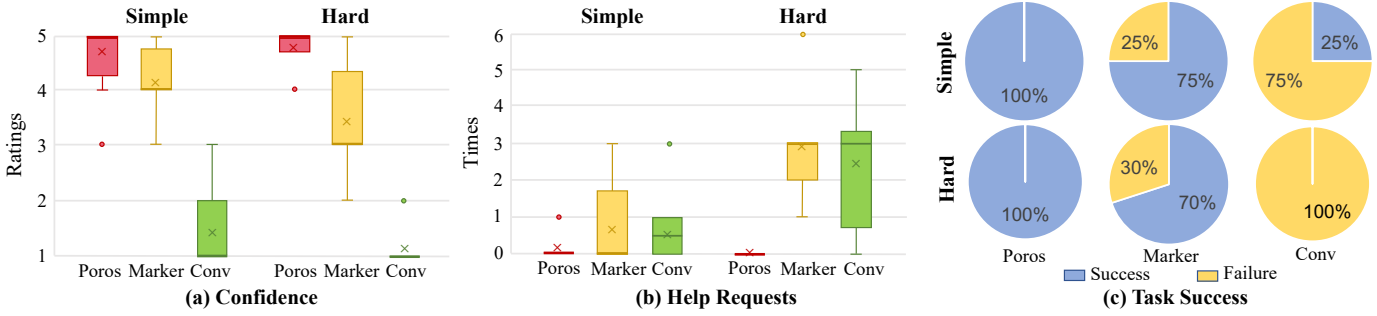


Figure 6: **Results on subjective confidence ratings, help requests, and task success.** (a) Participants' confidence ratings for Poros, Marker, and Conventional (Conv) approaches under Simple and Hard tasks. (b) Number of help requests recorded for each approach under Simple and Hard tasks. (c) Task success rates for the three approaches under both task levels.

This may be because Simple tasks involve fewer steps, allowing participants to rely on tactile markers to locate buttons. For Hard tasks, Poros received significantly higher ratings than both Conventional ($Z = -2.92, p < .017$) and Marker ($Z = -2.62, p < .017$).

These observations were consistent with participants' feedback. One participant noted, "Without any assistance, I'm not sure what the beeps of the microwave mean." With tactile markers, P13 reported, "I am confident about which button I am pressing because I remember the button layout. However, I still need to count how many times I press the button, especially when the task is complex." Overall, Poros received high operational confidence ratings, with more than 93% of participants giving ratings above 4. As P08 stated, "The speech guidance gives me peace of mind, even if something goes wrong, it will help me to correct it in time." However, P03 expressed less complete trust in Poros, noting that "because any device or system would not always be stable. I worry about whether the speech guidance is accurate." Such feedback gives us insights into BLV people's trust in tech products.

Help Requests. We recorded the number of help requests, as shown in Fig. 6(b). For Simple tasks, the median was $Md = 0$ for Poros, $Md = 0$ for Marker, and $Md = 0.5$ for Conventional. For Hard tasks, Poros again resulted in no help requests ($Md = 0$), whereas Marker ($Md = 3$) and Conventional ($Md = 3$) required substantially more requests. To examine whether these differences were statistically significant, we conducted Friedman tests on help request counts across the three approaches for Simple and Hard tasks separately. The results showed significant differences among the approaches in both Simple ($\chi^2(2) = 6.33, p < .05$) and Hard ($\chi^2(2) = 14.11, p < .05$) tasks. We further conducted Wilcoxon signed-rank tests with Bonferroni correction ($\alpha = .017$) as post-hoc analyses. For Simple tasks, no significant difference was found between Marker and Poros. For Hard tasks, Marker resulted in significantly more help requests than Poros ($Z = -2.87, p < .017$), and Conventional required significantly more requests than Poros ($Z = -2.55, p < .017$). This may be because Simple tasks involve fewer steps and are easier for participants to remember. Although participants rated Marker and Poros as similarly convenient (Fig. 5(a)), Marker still required more help requests on average. A Pearson correlation analysis revealed a negative correlation between average confidence ratings and help request counts ($r = -.522, p < .05$), suggesting that participants who felt more confident required less assistance during task execution.

Participants' subjective feedback further corroborated these performance patterns. As P09 noted, "The marker helped me locate the buttons, but it cannot tell me exactly what to do, such as how many times I should press a button or what the next step is. So I chose to ask for help." In contrast, without assistance for button localization and microwave status feedback, participants often stopped attempting the task halfway. As P01 explained, "I don't care about the next steps because I could not even locate the first button. I'm frustrated and choose to give up." When using Poros, help requests were rare because the system provided step-by-step guidance and state feedback during task execution. Only one help request occurred in the Simple tasks and none in the Hard tasks. This request occurred while Poros was assess-

ing the microwave state. During this interval, the participant hesitated and asked what to do next. Before the experimenter responded, Poros provided feedback indicating the current status and the next step.

Task Success. Figure 6(c) shows success rates across the three approaches and two task levels. Overall, Poros achieved the highest success rates, with all participants completing both Simple and Hard tasks. This reflects not only Poros's step-by-step guidance, but also its support for error recovery. For example, a participant might accidentally set 7 minutes instead of the target 5 minutes. Poros detected the mismatch between the displayed and expected states and provided corrective feedback to guide the participant back to the target setting. For the Marker approach, the success rate was 75% for Simple tasks and 70% for Hard tasks. Although tactile markers helped participants locate buttons, the lack of guidance on operation sequences and press counts still caused errors, especially in complex tasks. The Conventional approach resulted in very low success rates: only 25% of Simple tasks were completed successfully, and no Hard tasks were completed. Among failed Hard tasks, over 88% were abandoned midway. Participants attributed this to the difficulty of operating the microwave without assistance, and most gave up early without requesting help.

To examine the relationship between confidence and task performance, we computed Pearson correlations between average confidence ratings and task success rates across all approaches and task levels. The results showed a strong positive correlation ($r = .986, p < .05$), indicating that higher confidence was associated with higher success rates. Participants' feedback illustrated this phenomenon. In the Marker condition, some participants were surprised to learn they had completed a task successfully, especially for Hard tasks. As P02 explained: "This task is quite hard with many different buttons and different numbers of presses. I kept counting how many times I had pressed the button and felt I must have made some mistakes. So I was surprised by the result." In contrast, participants reported that Poros made operation easier and more reliable. As one participant noted, "Poros helps me not only complete the task successfully but also makes it possible to chat with the virtual agent while operating the microwave. I do not need to carefully check every step because Poros tells me what to do next."

5.3 Additional Post-Study Feedback

The post-study feedback captured participants' expectations and preferences across the three approaches, with detailed results provided in the supplementary material. All participants were dissatisfied with the default microwave design, with 100% rating it as "Not Meet Expectations." In contrast, Poros achieved the highest satisfaction, with 91.67% "Meet Expectations" for Simple tasks and 100% for Hard tasks. Participants reported that Poros simplified appliance interaction by explaining button layouts and functions, and that speech guidance increased confidence and supported multitasking. One participant noted, "Whether the task is simple or complex, Poros helps me operate the microwave easily." For the Marker approach, satisfaction varied by task complexity. For Simple tasks, 58.33% of participants rated it as "Meet Expectations" and 42.67% as "Somewhat Meet Expectations." For Hard tasks, only 40% rated it as "Meet Expectations," while 30% rated it as "Somewhat

Meet Expectations” and 30% as “Not Meet Expectations.” Participants considered tactile markers useful for routine tasks with simple, frequently used functions, but preferred Poros for new appliances or unfamiliar, complex tasks. As P09 and P13 noted, *“The cost of using markers to learn and remember complex functions is high. We could only touch or taste the food to find out if something went wrong.”*

6 DISCUSSION

Our study shows that Poros helps BLV users complete complex appliance operations through perception-enabled interactive assistance. Prior assistive technologies have improved accessibility through tactile augmentation [18, 23], smart device control [32, 50], and visual and environmental perception via interface recognition, scene descriptions, and multimodal feedback [8, 10, 15, 52]. While these systems improve access to visual information, Poros targets appliance operation workflows by supporting control localization, appliance-state interpretation, multi-step guidance, and error recovery. Our findings show that Poros helped participants operate the appliance with confidence, complete tasks successfully, and rely on minimal help, especially in complex, multi-step operations. These results motivate our discussion of design implications, applications, limitations, and future work.

6.1 Design Implications for Assistive Systems

Participants’ feedback and prior findings on context-dependent user behavior [40] highlight that appliance-assistance systems should adapt to different tasks, interfaces, and user preferences. We discuss three design implications for future appliance-assistance systems.

Assistive systems should balance minimal assistance with timely intervention based on users’ interaction state. Our results suggest that more guidance is not always necessary. For simple, familiar operations, tactile markers or prior knowledge may be sufficient when the main barrier is button localization. However, when users hesitate, repeatedly explore the panel, make mistakes, or encounter mismatched appliance states, minimal cues may no longer be enough. Poros addressed these situations through procedural guidance, state feedback, and corrective instructions. Future assistive systems should infer when help is needed and intervene accordingly, while avoiding unnecessary step-by-step guidance when users can proceed independently.

Assistive systems should provide feedback that matches different moments in appliance operation. Feedback should not rely on a single speech channel. Before a button press, spatial or differentiated auditory cues may help users approach the target control without lengthy verbal descriptions. After a press, a brief confirmation sound can indicate that the input has been recognized. When the appliance state changes, or when the displayed state does not match the intended target, speech feedback is more appropriate because it can explain the current state, the expected state, and the next corrective step. Future systems allow users to combine speech guidance, confirmation sounds, and spatial auditory cues according to task complexity and personal preference.

Assistive systems should support personality and practical deployment. Consistent with prior work on user-adaptive interaction [39], users may have different preferences for exploring appliance panels and activating controls. Some may use one or both hands to explore the panel, while others may prefer a finger or hand for button pressing. Assistive systems should avoid assuming a fixed posture, operating finger, or interaction sequence, and instead support transitions between exploration and activation. Future systems should consider lighter, more portable form factors so that perception, hand tracking, and speech feedback can support appliance use.

6.2 Applications

Different assistive approaches can complement each other in supporting appliance accessibility. The findings show that participants used strategies such as tactile markers or sighted demonstrations. Poros can complement these strategies by helping users understand operation procedures and confirm appliance states while using markers to locate familiar buttons, increasing confidence and reducing errors. In addition, when using a new appliance, Poros can serve as a step-by-step guide, similar to assistance from a sighted person. Poros could also

support access to public facilities, such as vending machines, ATMs, and self-service ticket machines. Integrated with navigation systems, Poros could help BLV users locate facilities, perceive their positions through audio cues, and receive step-by-step operational guidance. This combination of navigation and interaction support may increase users’ confidence and willingness to use unfamiliar public facilities. Beyond BLV users, Poros may also support sighted users and older adults who are unfamiliar with complex appliances. Instead of relying solely on textual manuals, users could receive interactive, spatially grounded guidance for learning appliance operations. For sighted users, a concise mode could provide procedural guidance without detailed spatial cues.

6.3 Limitation and Future Works

Poros targets household appliances with structured, multi-step procedures described in manuals, transforming these procedures into executable, state-based guidance. However, there are several limitations concerning the current scope and scalability of Poros.

First, adapting Poros to a new appliance does not require retraining or fine-tuning the LLM-based pipeline, but still requires appliance-specific preparation, including manual acquisition, model construction, control-panel annotation, HoloLens registration, and rule validation. Once validated, this configuration can be reused across users and tasks for the same appliance model or similar appliances with comparable operation rules, such as time and temperature adjustment. However, this preparation limits scalability when supporting many appliance models. Future work should reduce this effort by using multimodal LLMs and multi-view visual reasoning to automate manual parsing, control-panel annotation, panel registration, and rule validation.

Second, although appliance-specific elements such as button layout, panel registration, and state recognition are modular, the current prototype relies on appliance-specific perception modules. The display-state recognition pipeline is tailored to observable LED or digit-like indicators, such as time or power settings on microwave displays, which may limit applicability to appliances with other feedback mechanisms. Text displays, icons, touchscreens, or status indicators may require additional modules. Future implementations could use multimodal perception models, such as vision-language models, to interpret appliance feedback and map it to the states required by the instruction generator.

Third, our empirical evaluation is limited in application scope. Although our appliance set included different types of microwave ovens and microwave-combination appliances, the user study was conducted on a commonly used microwave oven. Poros may extend to appliances with similar structured procedures, control layouts, and visually observable display states, but other categories may introduce new challenges in interface layout, feedback mechanisms, and task structure. Future studies should evaluate Poros across broader appliance categories.

7 CONCLUSION

In this work, we presented Poros, a perception-enabled interactive assistance system that automatically models appliance usage logic from instruction manuals and provides real-time operational guidance for BLV users. By combining a constructed task state machine with real-time observations from integrated AR sensing, Poros enables BLV users to understand appliance states, locate controls, and follow step-by-step instructions during task execution. A comparative user study with quantitative and qualitative analysis shows that Poros improves appliance accessibility by enhancing task efficiency, success rates, and user confidence, while reducing the need for assistance, particularly for complex tasks. Post-study interviews indicated that Poros enhances current assistive strategies and is especially effective for learning new appliances and complex tasks. Our findings further suggest broader applications and design implications, such as adaptive guidance levels, flexible auditory feedback, and diverse interaction styles. Together, these insights suggest promising directions for developing more accessible and flexible appliance assistance systems in everyday environments.

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